Name

Mathematics Advanced Year 11

Algebraic Techniques

Book 1

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MAV-11-01 applies algebraic techniques and the laws of indices and surds to manipulate expressions and solve problems

Algebraic techniques

- Use index laws to simplify expressions and solve problems involving positive, negative, zero or fractional indices
- Expand, factorise and simplify algebraic expressions
- Simplify expressions involving algebraic fractions
- Expand and simplify expressions involving surds
- Identify the conjugate of $\sqrt{a} \pm \sqrt{b}$, and rationalise the denominators of expressions of the form $\frac{a}{\sqrt{b}\pm\sqrt{c}}$ and $\frac{\sqrt{a}\pm\sqrt{d}}{\sqrt{b}\pm\sqrt{c}}$, where a, b, c and d are positive rational numbers
- Solve quadratic equations $ax^2 + bx + c = 0$ by factorisation, completing the square and using the quadratic formula $x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$ where a, b and c are real numbers and $a \neq 0$
- Define the discriminant as $\Delta = b^2 - 4ac$ and use it to solve problems involving the number of real roots of a quadratic equation and determine the conditions for roots to be equal, distinct, real or rational

Basic Index Laws

Basic Index Laws

Multiplication	$a^m \times a^n = a^{m+n}$
Division	$\frac{a^m}{a^n} = a^{m-n}$
Power of a Power	$(a^m)^n = a^{m \times n}$
Zero Power	$a^0 = 1$
Power of a Product	$(ab)^m = a^m b^m$
Power of a Quotient	$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$

Example Apply index laws to simplify expressions with more than one variable

$$3x^2y^4 \times 2x^3y$$
$$= 6x^5y^5$$

$$\frac{9x^5y^2}{6x^2y^7}$$
$$= \frac{3x^3}{2y^5}$$

Guided Practice

Simplify

a $6x^3y^7 \times 2x^5y^2$

$12x^8y^9$

b $4c^4d \times 2c^2$

$8c^6d$

c $\frac{12x^2y^{11}}{8x^6y}$

$\frac{3y^{10}}{2x^4}$

d $\frac{2x^5y^3}{14x^6y}$

$\frac{y^2}{7x}$

Example Simplify expressions using power of a product

Simplify

$(5b)^3$

$= 5^3b^3$

$= 125b^3$

$(-3x^4y)^2$

$= (-3)^2x^8y^2$

$= 9x^8y^2$

$4x(c^2d^3)^5$

$= 4xc^{10}d^{15}$

Guided Practice

Simplify

a $(2x)^4$

$16x^4$

b $(3y^2)^3$

$27y^6$

c $(-4a)^2$

$16a^2$

d $(2a^2b)^3$

$8a^6b^3$

e $3(x^2)^4$

$3x^8$

f Compare $(3x)^2$ and $3x^2$

$9x^2, 3x^2$

Example Simplify expressions using power of a quotient

Simplify

$\left(\frac{6}{b}\right)^3 = \frac{6^3}{b^3}$

$= \frac{216}{b^3}$

$\left(\frac{ab^2}{c^4}\right)^4 = \frac{a^4b^8}{c^{16}}$

$\left(\frac{-2}{3b}\right)^4 = \frac{(-2)^4}{3^4b^4}$

$= \frac{16}{81b^4}$

Guided Practice

Simplify

a $\left(\frac{p}{q}\right)^2$

$\frac{p^2}{q^2}$

b $\left(\frac{4}{y}\right)^3$

$\frac{64}{y^3}$

c $\left(\frac{-5}{p^2}\right)^4$

$\frac{625}{p^8}$

d $\left(\frac{2x}{y}\right)^5$

$\frac{32x^5}{y^5}$

e $3\left(\frac{n^3}{2m}\right)^2$

$\frac{3n^6}{4m^2}$

f $\left(\frac{-3f}{2g^2}\right)^3$

$\frac{-27f^3}{8g^6}$

Negative Indices

ⓘ Index Laws of Negative Powers

$a^{-n} = \frac{1}{a^n}$	$\frac{1}{a^{-n}} = a^n$	$\left(\frac{a}{b}\right)^{-n} = \frac{b^n}{a^n}$

🔍 **Rewrite** expressions involving negative indices with positive indices

x^{-2}	$3x^{-2}$	$\frac{x^{-2}}{3}$	$12yx^{-7}$
$\frac{1}{x^2}$	$\frac{3}{x^2}$	$\frac{1}{3x^2}$	$\frac{12y}{x^7}$

Express the following with positive indices

a x^{-3}	**b** x^{-1}	**c** a^{-4}
d $3m^{-5}$	**e** p^7q^{-2}	**f** $x^{-4}y^{-3}$

🔍 **Rewrite** expressions involving negative indices with positive indices

$\frac{1}{x^{-2}}$	$\frac{3}{x^{-2}}$	$\frac{1}{3x^{-2}}$	$\frac{12}{yx^{-7}}$
x^2	$3x^2$	$\frac{x^2}{3}$	$\frac{12x^7}{y}$

Express the following with positive indices

a $\frac{1}{y^{-3}}$	**b** $\frac{3}{y^{-2}}$	**c** $\frac{1}{7x^{-8}}$
d $\frac{a^2}{b^{-3}}$	**e** $\frac{7}{x^{-4}y^3}$	**f** $\frac{3a}{y^{-5}}$

Example Simplify expressions using the reciprocal

$\left(\frac{x}{y^2}\right)^{-1}$

$= \frac{y^2}{x}$

$\left(\frac{x}{y}\right)^{-3}$

$= \left(\frac{y}{x}\right)^{3}$

$= \frac{y^3}{x^3}$

Guided Practice

Express the following with positive indices.

a $\left(\frac{5}{y}\right)^{-1}$

b $\left(\frac{x}{3}\right)^{-2}$

c $\left(\frac{x^3}{y}\right)^{-2}$

d $\left(\frac{3y}{x^5}\right)^{-1}$

e $\left(\frac{2x^5}{y}\right)^{-3}$

f $\left(\frac{1}{3}\right)^{-2}$

Fractional Indices

Fractional Indices

$$a^{\frac{m}{n}} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m$$

Example Convert expressions in index form to surds

$x^{\frac{2}{3}}$ $= \sqrt[3]{x^2}$	$x^{\frac{5}{7}}$ $= \sqrt[7]{x^5}$	$x^{-\frac{5}{7}}$ $= \frac{1}{\sqrt[7]{x^5}}$

Guided Practice

Express in surd form.

a $x^{\frac{2}{5}}$	**b** $x^{\frac{5}{2}}$	**c** $y^{\frac{2}{7}}$
d $9^{-\frac{3}{2}}$	**e** $y^{-\frac{2}{3}}$	**f** $\frac{a}{b^{\frac{3}{2}}}$

Example Convert expressions in surd form to index form

$\sqrt[3]{a^2}$ $= a^{\frac{2}{3}}$	$a\sqrt{a^3}$ $= a \times a^{\frac{3}{2}}$ $= a^{\frac{5}{2}}$	$\frac{y}{\sqrt[4]{x^7}}$ $= yx^{-\frac{7}{4}}$

Guided Practice

Express in index form.

a $\sqrt{x^3}$	**b** $\sqrt[5]{x^2}$	**c** $\sqrt{2a}$
d $x\sqrt{x^5}$	**e** $y\sqrt[3]{x^2}$	**f** $\frac{1}{\sqrt[3]{y^2}}$

Factorising Expressions with Indices

Factorising Expressions with Indices

- We can split a base based on prime factors.
  This allows us to simplify when the bases aren't the same.

Example Simplify expressions with indices by factorisation

6×2^n

$= 2 \times 3 \times 2^n$

$= 2^{n+1} \times 3$

$12^n \times 9 = (4 \times 3)^n \times 3^2$

$= 4^n \times 3^n \times 3^2$

$= 2^{2n} \times 3^{n+2}$

Guided Practice

Simplify by writing these expressions with prime bases.

a $2^n \times 4^n \times 8^n$

2^{6n}

b $18^n \times 12$

$2^{n+2} \times 3^{2n+1}$

c $100^{2n-1} \times 25^{-1} \times 8^{-1}$

$2^{4n-5} \times 5^{4n-4}$

d $\frac{9^{n+2} \times 3^{n+1}}{3 \times 27^n}$

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Factorising Expressions with Indices

- We can factorise expressions involving addition and subtraction by taking out a common factor. This is useful when simplifying fractions.

Example Simplify expressions with indices by factorisation

$5^n + 5^{n+2}$

$= 5^n(1 + 5^2)$

$= 5^n \times 26$

$3^n + 3^{n+1}$

$= 3^n + 3 \times 3^n$

$= 3^n(1 + 3)$

$= 4 \times 3^n$

Guided Practice

Simplify by factorising.

a $7^{n+2} + 7^n$

50×7^n

b $4^n + 12^n$

$4^n(1 + 3^n)$ or $2^{2n}(1 + 3^n)$

c $2^{2n+2} - 2^{2n-1}$

$7 \times 2^{2n-1}$ or 3.5×2^{2n}

d $\frac{3^{n+3} - 3^n}{3^n}$

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Expanding Expressions

Expanding Brackets

To expand brackets, we use the distributive law:

$$a(b+c) = ab + ac$$

After expanding brackets, **simplify** the result by combining **like terms**.

Example Expand then simplify the result

$4(3p+2) - 5p$ $= 12p + 8 - 5p$ $= 7p + 8$	$5(2x-3) - 4(1-5x)$ $= 10x - 15 - 4 + 20x$ $= 30x - 19$

Guided Practice

Expand, then simplify.

a $5(3m+2) + 4m$

$= 15m + 10 + 4m$

$=$

$19m + 10$

b $3(1-5e) + 6e$

$3 - 9e$

c $4w - 2(5+2w)$

$= 4w - 10 - 4w$

$=$

-10

d $8 - 5(2x-7)$

$43 - 10x$

e $5(2a+3) + 4(a+7)$

$14a + 43$

f $3(2d+3) - 5(d+4)$

$d - 11$

Expand Binomial Products using the Rule

1. Multiply each term in the first bracket by each term in the second bracket.
2. Simplify the result.

$$(a+b)(c+d)$$
$$= a(c+d) + b(c+d)$$
$$= ac + ad + bc + bd$$

Shortcut:

$$(a+b)(c+d) = ac + ad + bc + bd$$

Example Expand binomial products

$(x+3)(x+5)$

$$(x+3)(x+5) = x^2 + 5x + 3x + 15$$
$$= x^2 + 8x + 15$$

$(x-4)(x+7)$

$$(x-4)(x+7) = x^2 + 7x - 4x - 28$$
$$= x^2 + 3x - 28$$

Guided Practice

Expand and simplify these expressions.

a $(x+4)(x+2)$

$x^2 + 6x + 8$

b $(x-5)(x-2)$

$x^2 - 7x + 10$

c $(2x-1)(x-6)$

$2x^2 - 13x + 6$

d $(5x-2)(3x+7)$

$15x^2 + 29x - 14$

ⓘ Perfect Square Expansion

$$(a+b)^2 = a^2 + 2ab + b^2$$

1. Identify a and b.
2. Write the expression $a^2 + 2ab + b^2$ using a and b from step 1.
3. Simplify.

Example Expand perfect squares

$(x+4)^2$

1. $a = x \quad b = 4$
2. $(x)^2 + 2(x)(4) + (4)^2$
3. $x^2 + 8x + 16$

$(5x-3y)^2$

1. $a = 5x \quad b = -3y$
2. $(5x)^2 + 2(5x)(-3y) + (-3y)^2$
3. $25x^2 - 30xy + 9y^2$

Guided Practice

Expand these expressions

a $(x+y)^2$

$x^2 + 2xy + y^2$

b $(x+3)^2$

$x^2 + 6x + 9$

c $(x-7)^2$

$x^2 - 14x + 49$

d $(3x+1)^2$

$9x^2 + 6x + 1$

e $(6x-5)^2$

$36x^2 - 60x + 25$

f $(8y+3x)^2$

$64y^2 + 48xy + 9x^2$

Difference of Two Squares

$$(a-b)(a+b)=a^2-b^2$$

1. Identify a and b.
2. Write the expression a^2-b^2 using a and b from step 1.
3. Simplify.

Example Expand difference of two squares

$(y-3)(y+3)$	$(6-2t)(6+2t)$
1. $a=y \quad b=3$	1. $a=6 \quad b=2t$
2. $(y)^2-(3)^2$	2. $(6)^2-(2t)^2$
3. y^2-9	3. $36-4t^2$

Guided Practice

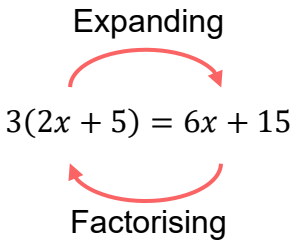
Expand these expressions

a $(x-4)(x+4)$ x^2-16	**b** $(x+8)(x-8)$ x^2-64
c $(2x+3)(2x-3)$ $4x^2-9$	**d** $(7-3x)(7+3x)$ $49-9x^2$
e $(x^3+5)(x^3-5)$ x^6-25	**f** $(3x^2-5)(3x^2+5)$ $9x^4-25$

Factorising Expressions

ⓘ Factorising

Factorising is the **inverse** of expanding.



To factorise:

1. Identify the highest common factor (HCF) of the terms. Place it outside brackets.
2. Divide each term by the HCF. Put the results inside the brackets.

Example Factorise expressions (algebraic HCF)

Factorise

$ab - 5b$	$8xy + 12y$	$10a^2 - 15a$
1. $= b(\quad)$	1. $= 4y(\quad)$	1. $= 5a(\quad)$
2. $= b(a - 5)$	2. $= 4y(2x + 3)$	2. $= 5a(2a - 3)$

Guided Practice

Factorise

a $xk + xc$	**b** $x^2 + 3x$
c $x^2 - 3x$	**d** $6xb + 8b$
e $3ab - 5b$	**f** $4xy^2 - 6y^2$

ⓘ Factorising a Negative HCF

To **factorise** with a **negative HCF**:

1. Identify the negative **highest common factor** (HCF) of the terms.
   Write it outside brackets.
2. Divide each term by the HCF. Put the results inside the brackets.

You should factor out a negative HCF if both terms are negative, or if the question asks for it.
This skill is used in a later topic (factorising by grouping in pairs).

Example Factorise by finding a finding a negative HCF

Factorise these expressions by finding a negative HCF.

$-8x - 16$	$12x - 4xy$
1. $-8(\quad)$	1. $-4x(\quad)$
2. $-8(x + 2)$	2. $-4x(-3 + y)$

Guided Practice

Factorise these expressions by finding a negative HCF.

a $-8x - 4$ $= -4(\quad)$ $-4(2x + 1)$	**b** $-2m + 4n$ $= -2(\quad)$ $-2(m - 2n)$	**c** $-3xy + 18x$ $-3x(y - 6)$
d $8x^2 - 12x$ $-4x(-2x + 3)$	**e** $6x - 20x^2$ $-2x(-3 + 10x)$	**f** $5a^3x + 10a^2$ $-5a^2(-ax - 2)$

Factorising Monic Quadratic Trinomials

To factorise a monic quadratic trinomial: $x^2 + bx + c = 0$:

1. Identify the product: c
2. Identify the sum: b
3. Think of a factor pair, α and β, that has a product of c and sum to b.
4. Factorise $x^2 + bx + c = (x + \alpha)(x + \beta)$.

Example Factorise monic quadratic trinomials

$x^2 - 13x + 36$

4. $(x - 4)(x - 9)$

1. P: 36
2. S: -13
3. F: $-4, -9$

$a^2 + 12a - 28$

4. $(a + 14)(a - 2)$

1. P: -28
2. S: 12
3. F: $14, -2$

Guided Practice

Factorise these expressions.

a $a^2 + 3a + 2$

$(a \quad)(a \quad)$

P: 2
S: 3
F: ……

$(a + 2)(a + 1)$

b $y^2 + 9y + 20$

P: ……
S: ……
F: ……

$(y + 4)(y + 5)$

c $x^2 - 7x + 12$

P: 12
S: -7
F: ……

$(x - 3)(x - 4)$

d $w^2 - 2w - 8$

P: ……
S: ……
F: ……

$(w - 4)(w + 2)$

e $y^2 - 10y + 21$

P: ……
S: ……
F: ……

$(y - 7)(y - 3)$

f $x^2 - 5x - 24$

P: ……
S: ……
F: ……

$(x - 8)(x + 3)$

Factorising Perfect Square Trinomials

$a^2 + 2ab + b^2$	$a^2 - 2ab + b^2$
$= (a + b)^2$	$= (a - b)^2$

1. Find the square roots of the first term $\sqrt{a^2} = a$ and last term $\sqrt{b^2} = b$.
2. Check whether the middle term is $2ab$.
3. Write $(a + b)^2$ or $(a - b)^2$ based on the sign of the middle term.

Example Factorise perfect square trinomials

$x^2 - 6x + 9$	$9x^2 + 30x + 25$
1. $\sqrt{x^2} = x$ $\quad \sqrt{9} = 3$	1. $\sqrt{9x^2} = 3x$ $\quad \sqrt{25} = 5$
2. Check: $2 \times x \times 3 = 6x$ ✓	2. Check: $2 \times 3x \times 5 = 30x$ ✓
3. $= (x - 3)^2$	3. $= (3x + 5)^2$

Guided Practice

Factorise these expressions

a $x^2 + 10x + 25$ $(x + 5)^2$	**b** $x^2 + 18x + 81$ $(x + 9)^2$
c $p^2 + 2pq + q^2$ $(p + q)^2$	**d** $x^2 - 16x + 64$ $(x - 8)^2$
e $25x^2 + 60x + 36$ $(5x + 6)^2$	**f** $x^6 - 14x^3 + 49$ $(x^3 - 7)^2$

Factorising Difference of Two Squares

$$a^2 - b^2$$
$$= (a - b)(a + b)$$

1. Find the square roots of each term $\sqrt{a^2} = a$ and $\sqrt{b^2} = b$.
2. Write $(a - b)(a + b)$.

Example Factorise difference of two squares

$a^2 - 36$

1. $\sqrt{a^2} = a$ $\quad \sqrt{36} = 6$
2. $(a - 6)(a + 6)$

$25x^6 - 9x^4$

1. $\sqrt{25x^6} = 5x^3$ $\quad \sqrt{9x^4} = 3x^2$
2. $(5x^3 - 3x^2)(5x^2 + 3x^2)$

Guided Practice

Factorise these expressions

a $x^2 - 1$

$(x - 1)(x + 1)$

b $25 - x^2$

$(5 - x)(5 + x)$

c $4x^2 - 9$

$(2x - 3)(2x + 3)$

d $49x^2 - 4y^2$

$(7x - 2y)(7x + 2y)$

e $a^2 - \frac{1}{9}$

$\left(a - \frac{1}{3}\right)\left(a + \frac{1}{3}\right)$

f $\frac{y^2}{16} - \frac{4}{9}$

$\left(\frac{y}{4} - \frac{2}{3}\right)\left(\frac{y}{4} + \frac{2}{3}\right)$

Factorising by Noticing Binomial Common Factors

$$a(c+d)+b(c+d)$$
$$=(c+d)(a+b)$$

1. Identify the **binomial** that divides into all terms. Write it outside a second set of brackets.
2. Divide each term by the binomial HCF. Write the results inside a second set of brackets.

Example Factorise by noticing the binomial HCF

$3(x+y)+x(x+y)$

1. $(x+y)(\quad)$
2. $(x+y)(3+x)$

$5(2x-y)-3x(2x-y)$

1. $(2x-y)(\quad)$
2. $(2x-y)(5-3x)$

Guided Practice

Factorise.

a $4(x+3)+x(x+3)$

$=(x+3)(\quad)$

$(x+3)(4+x)$

b $3(x+1)-x(x+1)$

$(x+1)(3-x)$

c $x(x-7)+2(x-7)$

$(x-7)(x+2)$

d $8(a-4)-a(a-4)$

$(a-4)(8-a)$

e $x(7+3x)+(7+3x)$

$(7+3x)(x+1)$

f $y(2x-5)-(2x-5)$

$(2x-5)(y-1)$

Factorising by Noticing Binomial Common Factors

Sometimes the binomial common factor is not obvious.

1. Factorise a pair of terms such that the expression is inside brackets.
2. Factorise by noticing the binomial HCF.

Example Factorise by noticing binomial HCF after factorising one pair of terms

$7 - 2x + x(7 - 2x)$

1. $1(7 - 2x) + x(7 - 2x)$
2. $(7 - 2x)(1 + x)$

$7x(x - 4) - x + 4$

1. $7x(x - 4) - (x - 4)$
2. $(x - 4)(7x - 1)$

Guided Practice

Factorise these expressions.

a $7 + 3x + x(7 + 3x)$

$= 1(7 + 3x) + x(7 + 3x)$

$=$

$(7 + 3x)(1 + x)$

b $y(4y - 1) - (4y - 1)$

$(4y - 1)(y - 1)$

c $y(2x - 5) - 2x + 5$

$(2x - 5)(y - 1)$

d $2x(x + 1) - x - 1$

$(x + 1)(2x - 1)$

e $5y(x - 7) - x + 7$

$(x - 7)(5y - 1)$

f $12x^2(y - 3) + 3 - y$

$(y - 3)(12x^2 - 1)$

Factorise by Grouping in Pairs

$$ac + ad + bc + bd$$

Factorise expressions with **four terms** with no common factor using **grouping in pairs**:

1. Pair the terms.
2. Factorise each pair separately.
3. Factorise using the binomial common factor.

⚠ You may need to take out a negative factor to get a binomial common factor.

Example Factorise by grouping in pairs

$6xy + 9x + 4y + 6$

1. $6xy + 9x + 4y + 6$
2. $3x(2y + 3) + 2(2y + 3)$
3. $(2y + 3)(3x + 2)$

The first step is to ..... up the first two terms and the last two terms.

From the first pair, factor out .....

From the second pair, factor out .....

The binomial common factor is .....

Guided Practice

Factorise:

a $2x + 8 + bx + 4b$

$(x + 4)(2 + b)$

b $m^2 - 2m + 3m - 6$

$(m - 2)(m + 3)$

c $3x^2 + 6x - 2x - 4$

$(x + 2)(3x - 2)$

d $2ax + bx - 2ay - by$

$(2a + b)(x - y)$

Example Factorise by grouping in pairs

$2x - 4 + 6y - 3xy$

1. $2x - 4 + 6y - 3xy$
2. $2(x - 2) + 3y(2 - x)$
3. $2(x - 2) - 3y(x - 2)$

$(x - 2)(2 - 3y)$

In this example, we needed to factor out a ..... from $3y(2 - x)$ to make ..... so that we could get a c..... f.....

Guided Practice

Factorise:

a $6x - 3 + 5y - 10xy$

$= 3(2x - 1) + 5y(1 - 2x)$

$=$

$(2x - 1)(3 - 5y)$

b $3x - 15 + 5y - xy$

$(x - 5)(3 - y)$

c $6 - 2a + a^3 - 3a^2$

$(3 - a)(2 - a^2)$

d $2ax - bx + by - 2ay$

$(2a - b)(x - y)$

ⓘ Factorising Non-Monic Quadratic Trinomials

Non-monic means the coefficient of x^2 is not 1.
Non-monic quadratic trinomials are of the form $ax^2 + bx + c$.

To factorise a non-monic quadratic trinomial: $ax^2 + bx + c = 0$

1. Identify the product: ac
2. Identify the sum: b
3. Find a factor pair, α and β, that has product of ac and sum to b.
4. Split bx to $\alpha x + \beta x$
5. Factorise by grouping in pairs.

Example Factorise non-monic quadratic trinomials

$$5x^2 - 13x + 6$$

1. P: $5 \times 6 = 30$
2. S: -13
3. F: $10, -3$
4. Split $= 5x^2 - 10x - 3x + 6$
5. Factorise $= 5x(x - 2) - 3(x - 2)$
   $= (x - 2)(5x - 3)$

Guided Practice

Factorise:

a $3x^2 + 4x + 1$

P: 3
S: 4
F: .....

$(3x + 1)(x + 1)$

b $2x^2 + 11x + 5$

P: .....
S: .....
F: .....

$(2x + 1)(x + 5)$

c $6x^2 + 5x - 6$

P: .....
S: .....
F: .....

$(3x - 2)(2x + 3)$

d $2x^2 - 7x - 15$

P: .....
S: .....
F: .....

$(2x + 3)(x - 5)$

Basic Algebraic Fractions

Producing Equivalent Algebraic Fractions

Produce equivalent fractions by multiplying the numerator and denominator by the same term.

Example Produce equivalent algebraic fractions

$$\frac{3n}{7} = \frac{\quad}{14}$$
$$= \frac{3n}{7} \frac{\times 2}{\times 2}$$
$$= \frac{6n}{14}$$

$$\frac{3a}{4b^2} = \frac{\quad}{12b^7}$$
$$= \frac{3a}{4b^2} \frac{\times 3b^5}{\times 3b^5}$$
$$= \frac{9ab^5}{12b^7}$$

Guided Practice

Produce equivalent fractions.

a

$$\frac{2y}{5} = \frac{\quad}{15}$$

$6y$

b

$$\frac{5a}{2} = \frac{\quad}{12}$$

$30a$

c

$$\frac{3t}{4w} = \frac{\quad}{16w}$$

$12t$

d

$$\frac{6x}{5y} = \frac{\quad}{5yz}$$

$6xz$

e

$$\frac{7m}{9n} = \frac{\quad}{18np}$$

$14mp$

f

$$\frac{5a}{4bc} = \frac{\quad}{8b^2c^3}$$

$10abc^2$

Simplifying Fractions involving Indices

- To simplify fractions:

1. Divide the numerator and denominator by a common factor.

- When indices are involved, use the index law:

$$\frac{a^m}{a^n} = a^{m-n}$$

Example Simplify algebraic fractions involving indices

$$\frac{10a^5}{5a^2}$$

$$= \frac{10a^5}{5a^2} \begin{matrix} \div a^2 \\ \hline \div a^2 \end{matrix}$$

$$= \frac{10a^3}{5} \begin{matrix} \div 5 \\ \hline \div 5 \end{matrix}$$

$$= 2a^3$$

$$\frac{8ab}{2a^2}$$

$$= \frac{8ab}{2a^2} \begin{matrix} \div a \\ \hline \div a \end{matrix}$$

$$= \frac{8b}{2a} \begin{matrix} \div 2 \\ \hline \div 2 \end{matrix}$$

$$= \frac{4b}{a}$$

Guided Practice

Simplify these expressions.

a $\frac{35x^2}{7x}$ $5x$

b $\frac{-14x^3}{7x}$ $-2x^2$

c $-\frac{36ab^2}{4ab}$ $-9b$

d $\frac{8xy^3}{4x^2y}$ $\frac{2y^2}{x}$

e $\frac{-15pq^2}{30p^2q^2}$ $-\frac{1}{2p}$

f $\frac{-20s}{45s^2t}$ $-\frac{4}{9st}$

ⓘ Adding and Subtracting Algebraic Fractions

1. Rewrite the fractions with the lowest common multiple of the denominators.
2. Add or subtract the numerators.

Example Add/subtract algebraic fractions with pronumeral denominators

$$\frac{2}{5}+\frac{3}{a^2}$$

1. Same denominators $\quad \frac{\quad}{5a^2}+\frac{\quad}{5a^2}$

$$\frac{2a^2}{5a^2}+\frac{15}{5a^2}$$

2. Add numerators $\quad \frac{2a^2+15}{5a^2}$

$$\frac{2}{3x}-\frac{3}{5x}$$

1. Same denominators $\quad \frac{\quad}{15x}-\frac{\quad}{15x}$

$$\frac{10}{15x}-\frac{9}{15x}$$

2. Subtract numerators $\quad \frac{1}{15x}$

Guided Practice

Simplify these expressions.

a $\frac{1}{a}+\frac{1}{3a}$

$=\frac{\quad}{3a}+\frac{\quad}{3a}$

$=\frac{\quad}{3a}$

$\frac{4}{3a}$

b $\frac{1}{2x}-\frac{1}{3x}$

$=\frac{\quad}{6x}-\frac{\quad}{6x}$

$=$

$\frac{1}{6x}$

c $\frac{2}{a}-\frac{3}{4a}$

$\frac{5}{4a}$

d $\frac{3}{5x}-\frac{1}{2x}$

$\frac{1}{10x}$

e $\frac{5}{a}+\frac{2}{a^3}$

$\frac{5a^2+2}{a^3}$

f $\frac{1}{x^2}-\frac{1}{y}$

$\frac{y-x^2}{x^2y}$

Multiplying Algebraic Fractions

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

Example Add/subtract algebraic fractions with pronumeral denominators

$$\frac{2a}{5} \times \frac{3b}{7}$$
$$= \frac{2a \times 3b}{5 \times 7}$$
$$= \frac{6ab}{35}$$

$$\frac{4x}{15} \times \frac{3}{2x}$$
$$= \frac{4x \times 3}{15 \times 2x}$$
$$= \frac{12x}{30x} = \frac{2}{5}$$

Guided Practice

Simplify these expressions.

a $\frac{x}{3} \times \frac{2}{5}$

$\frac{2x}{15}$

b $\frac{2}{3} \times \frac{4a}{5}$

$\frac{8a}{15}$

c $\frac{6x}{5} \times \frac{7y}{6}$

$\frac{7xy}{5}$

d $\frac{8a}{5} \times \frac{3b}{4c}$

$\frac{6ab}{5c}$

e $\frac{3x}{2} \times 6$

$9x$

f $\frac{10x}{3} \times \frac{3}{10x}$

1

Algebraic Fractions with Indices

- When working with algebraic fractions with indices, it is helpful to write out every line of working out. Messy work leads to more mistakes.

Example Multiply and divide algebraic fractions involving indices

$$\frac{a^2b^4}{6} \times \frac{9}{a^2b^2}$$
$$= \frac{9a^2b^4}{6a^2b^2}$$
$$= \frac{9b^2}{6}$$
$$= \frac{3b^2}{2}$$

$$\frac{3x^2}{8y^5} \div \frac{15x^3}{4y}$$
$$= \frac{3x^2}{8y^5} \times \frac{4y}{15x^3}$$
$$= \frac{12x^2y}{120x^3y^5}$$
$$= \frac{12}{120xy^4}$$
$$= \frac{1}{10xy^4}$$

Guided Practice

Simplify these expressions.

a $\frac{x^4}{n^2} \times \frac{m}{n^3}$ $\frac{mx^4}{n^5}$

b $\frac{x}{y} \times \frac{y^2}{x}$ y

c $\frac{xz}{5} \div \frac{x}{10z}$ $2z^2$

d $\frac{2x}{5y} \times \frac{2y}{5x^2}$ $\frac{4}{25x}$

e $\frac{3}{4y^3x} \times \frac{3x^2}{y^2}$ $\frac{9x}{4y^5}$

f $\frac{3x^2}{5y^2} \div \frac{2x}{3y}$ $\frac{9x}{10y}$

Algebraic Fractions with Binomial Numerators

ⓘ Equivalent Fractions with Binomial Numerators

When creating equivalent algebraic fractions:

- Multiply both numerator and denominator by the same factor.
- Remember to multiply *each* term in the numerator by your factor.

$$\frac{a+b}{c} \times \frac{n}{n} = \frac{n(a+b)}{cn}$$

Example Produce equivalent fractions involving binomial numerator

$$\frac{4x+7}{3} = \frac{\quad}{15}$$
$$= \frac{4x+7}{3} \frac{\times 5}{\times 5}$$
$$= \frac{5(4x+7)}{15}$$
$$= \frac{20x+35}{15}$$

$$\frac{3x+9}{18} = \frac{\quad}{6}$$
$$= \frac{3x+9}{18} \frac{\div 3}{\div 3}$$
$$= \frac{x+3}{6}$$

Guided Practice

Determine the equivalent fraction.

a $\frac{2x+1}{3} = \frac{\quad}{12}$

$= \frac{2x+1}{3} \frac{\times 4}{\times 4}$

$\frac{8x+4}{12}$

b $\frac{5x-8}{2} = \frac{\quad}{6}$

$\frac{15x-24}{6}$

c $\frac{6x-8}{4} = \frac{\quad}{2}$

$\frac{3x-4}{2}$

d $\frac{12x+14}{6} = \frac{\quad}{3}$

$\frac{6x+7}{3}$

e $\frac{11x+2}{4} = \frac{\quad}{-12}$

$\frac{-33x-6}{-12}$

f $\frac{3x-20}{6} = \frac{\quad}{30}$

$\frac{15x-100}{30}$

Adding Fractions with Binomial Numerators

3. Rewrite the fractions with the **lowest common multiple** of the denominators.
4. Add the numerators.
5. Expand and simplify.

Example Produce equivalent fractions involving binomial numerator

$\frac{y+1}{5} + \frac{2y}{3}$

1. same denominators $\frac{3(y+1)}{15} + \frac{5(2y)}{15}$
2. add numerators $\frac{3(y+1)+5(2y)}{15}$
3. simplify $\frac{3y+3+10y}{15}$

$= \frac{13y+3}{15}$

$\frac{x+3}{2} + \frac{x-2}{5}$

1. same denominators $\frac{5(x+3)}{10} + \frac{2(x-2)}{10}$
2. add numerators $\frac{5(x+3)+2(x-2)}{10}$
3. simplify $\frac{5x+15+2x-4}{10}$

$= \frac{7x+11}{10}$

Guided Practice

Simplify these expressions.

a $\frac{x+6}{2} + \frac{x}{5}$

$= \frac{(\quad)}{10} + \frac{}{10}$

$\frac{7x+30}{10}$

b $\frac{x+3}{3} + \frac{x}{4}$

$= \frac{(\quad)}{12} + \frac{}{12}$

$\frac{7x+12}{12}$

c $\frac{2x+5}{5} + \frac{x}{10}$

$\frac{x+2}{2}$

d $\frac{x+1}{2} + \frac{x+3}{5}$

$\frac{7x+11}{10}$

e $\frac{y+4}{5} + \frac{y-3}{6}$

$\frac{11y+9}{30}$

f $\frac{2x-3}{6} + \frac{x+2}{8}$

$\frac{11x-6}{24}$

ⓘ Negative Fractions

A negative symbol in line with the fraction bar has several possible interpretations;
you can expand the negative into either the numerator or denominator, but not both.

$$-\frac{a}{b}=\frac{-a}{b}=\frac{a}{-b}$$

$$-\frac{a+b}{c}=\frac{-a-b}{c}=\frac{a+b}{-c}$$

🔍 Working with Negative Fractions

Write as many equivalent fractions as possible.

a $-\frac{3}{5}=$

b $-\frac{x+y}{3}=$

c $-\frac{x-y}{3}=$

d $\frac{x-y}{3}=$

e $-\frac{x+y}{a+b}=$

f $-\frac{x-y}{a+b}=$

g $-\frac{x+y}{a-b}=$

h $-\frac{x-y}{a-b}=$

True or false?

i $\frac{a+b}{c}=\frac{-a-b}{-c}$	**j** $\frac{a+b}{-c}=\frac{-a+b}{c}$	**k** $-\frac{a+b}{c}=\frac{-(a+b)}{c}$
l $-\frac{a-b}{c}=\frac{b-a}{c}$	**m** $-\frac{a+b}{c}=\frac{-a-b}{-c}$	**n** $\frac{a-b}{c-d}=\frac{b-a}{d-c}$

Subtracting Fractions with Binomial Numerators

1. Rewrite the fractions with the lowest common multiple of the denominators.
2. Subtract the numerator.
3. Expand and simplify.

Example Subtract algebraic fractions with binomial numerators

$$\frac{x-1}{3}-\frac{x+4}{5}$$

1. same denominators $\frac{5(x-1)}{15}-\frac{3(x+4)}{15}$
2. subtract $\frac{5(x-1)-3(x+4)}{15}$
3. simplify $\frac{5x-5-3x-12}{15}$

$$=\frac{2x-17}{15}$$

$$\frac{2x-3}{6}-\frac{3-x}{5}$$

1. same denominators $\frac{5(2x-3)}{30}-\frac{6(3-x)}{30}$
2. subtract $\frac{5(2x-3)-6(3-x)}{30}$
3. simplify $\frac{10x-15-18+6x}{30}$

$$=\frac{16x-33}{30}$$

Guided Practice

Simplify these expressions.

a $\frac{x-4}{2}-\frac{x+2}{3}$

$=\frac{3(\quad)}{6}-\frac{2(\quad)}{6}$

$=\frac{3(\quad)-2(\quad)}{6}$

$\frac{x-16}{6}$

b $\frac{x+1}{2}-\frac{2x+3}{5}$

$\frac{x-1}{10}$

c $\frac{2x+1}{4}-\frac{2-x}{3}$

$\frac{10x-5}{12}$

Identify whether a fraction can be simplified

A fraction $\frac{A}{B}$ can be simplified if and only if A and B share a common factor other than 1.

✗	✓	✓	✓	✗
$\frac{2x+3}{3}$	$\frac{4x+8}{2}$	$\frac{(x-3)(x+3)}{x+3}$	$\frac{(x+2)(x+3)}{x+2}$	$\frac{(x+2)(x+3)}{x-2}$

Decide whether these fractions can be simplified.

a $\frac{3x+9}{3}$	**b** $\frac{3x+5}{3}$	**c** $\frac{(x-4)(x+4)}{x+4}$
d $\frac{x^2+3}{x}$	**e** $\frac{x^2+3x}{x}$	**f** $\frac{(x+4)(x+2)}{x-2}$

ⓘ Simplifying Algebraic Fractions by Simple Factorisation

1. Factorise the numerator and denominator.
2. Cancel any common factors.

Example Simplify algebraic fractions by simple factorisation

$\frac{6x+8}{6}$

1. $\frac{2(3x+4)}{6}$
2. $\frac{3x+4}{3}$

We factorise to see if the numerator and denominator have common ……

$\frac{3x+4}{3}$ can't be simplified further because the numerator and denominator do not share any common ……

Guided Practice

Simplify

a $\frac{5a+10}{5}$

$= \frac{5(\quad)}{5}$

$=$

$a+2$

b $\frac{8}{4d-2}$

$= \frac{8}{2(\quad)}$

$=$

$\frac{4}{2d-1}$

c $\frac{2p+2q}{p+q}$

2

d $\frac{10}{2x^2+4x}$

$\frac{5}{x(x+2)}$

e $\frac{x^2-4}{x+2}$

$x-2$

f $\frac{9-x^2}{3-x}$

$3+x$

Identify whether numerator and denominator are negatives of each other

The numerator and denominator are negatives of each other when one can be obtained from the other by factorising out -1. In other words, all terms have the opposite sign.

✗	✓	✓	✓	✗
$\frac{x-3}{x+3}$	$\frac{-x+3}{x+3}$	$\frac{3-x}{x+3}$	$\frac{2x-7}{7-2x}$	$\frac{2x-7}{7+2x}$

Decide whether these numerators and denominators are negative of each other.

a $\frac{5-x}{x+5}$	**b** $\frac{5-x}{x-5}$	**c** $\frac{3x-1}{1-3x}$
d $\frac{3x-1}{1+3x}$	**e** $\frac{-1-3x}{3x+1}$	**f** $\frac{-5x+7}{5x-7}$

Example Simplify algebraic fractions by simple factorisation

$$\frac{3-x}{x-3}$$
$$= \frac{-(x-3)}{x-3}$$
$$= -1$$

$$\frac{9-x^2}{x-3}$$
$$= \frac{(3-x)(3+x)}{x-3}$$
$$= \frac{-(x-3)(3+x)}{x-3}$$

Guided Practice

Simplify

a $\frac{5-x}{x-5}$ -1	**b** $\frac{2x-7}{7-2x}$ -1	**c** $\frac{6-9x}{3x-2}$ -3
d $\frac{5-10x}{6x-3}$ $-\frac{5}{3}$	**e** $\frac{16-x^2}{x-4}$ $-(x+4)$	**f** $\frac{x^2-9}{3-x}$ $-(x+3)$

Simplify fractions by cancelling binomial factors

Simplify these fractions by cancelling common factors.

a $\frac{(x+1)(x+4)}{(x+1)}$	**b** $\frac{(x-3)(x+2)}{(x+2)(x+5)}$	**c** $\frac{2(x+3)(x-1)}{8(x+3)(x+2)}$
d $\frac{(x+3)(x+4)}{(x+4)}$	**e** $\frac{2(x-1)(x+5)}{4(x+5)}$	**f** $\frac{(x-3)(x+2)}{3(x+2)(x+7)}$
g $\frac{x-2}{(2-x)(x+3)}$	**h** $\frac{(x-5)(x+7)}{(x+5)(5-x)}$	**i** $\frac{3(x-4)^2}{(x-4)(x+2)}$

Simplify fractions by cancelling factors

Simplify these fractions by cancelling common factors.

a $\frac{(x+2)(x-3)}{(x+1)(x+2)} \times \frac{(x+1)(x+5)}{(x-3)(x+4)}$

b $\frac{3(x-1)}{(x+2)(x-1)} \times \frac{(x+2)(x+3)}{6}$

c $\frac{(x+4)(x-2)}{(x+3)(x+4)} \div \frac{(x-2)}{(x+3)(x+1)}$

d $\frac{(x+1)(x-2)}{(x+3)(x+1)} \times \frac{(x+3)(x+5)}{(x-2)(x+7)}$

e $\frac{2(x+4)}{(x-1)(x+4)} \div \frac{6}{(x-1)}$

f $\frac{(x+1)}{(1-x)} \div \frac{(x+5)}{(x-1)}$

Factorising Algebraic Fractions by Further Factorisation

1. Factorise the numerator and denominator.
2. Cancel any common factors.

Example Factorise Algebraic Fractions by Further Factorisation

$$\frac{x^2-x}{x^2+2x-3}$$

1. $\frac{x(x-1)}{(x+3)(x-1)}$
2. $\frac{x}{x+3}$

$$\frac{x^2-1}{2x^2+4x+2}$$

1. $\frac{(x-1)(x+1)}{2(x^2+2x+1)}$

   $\frac{(x-1)(x+1)}{2(x+1)^2}$
2. $\frac{x-1}{2(x+1)}$

Guided Practice

Simplify

a $\frac{x^2+5x+6}{x+2}$

$x+3$

b $\frac{x^2+6x+8}{x+4}$

$x+2$

c $\frac{a^2-9}{a^2+a-12}$

$\frac{a+3}{a+4}$

d $\frac{x^2+11x+28}{x^2+7x+12}$

$\frac{x+7}{x+3}$

Multiplying and Dividing Fractions by Factorising

0. **Change division to multiplication** by the reciprocal.
1. **Factorise** all numerators and denominators.
2. **Cancel** any common factors vertically or diagonally.
3. **Multiply** the numerators and multiply the denominators.

Example Multiply and divide algebraic fractions by factorising

$$\frac{2a}{a^2-9}\times\frac{a-3}{5a}$$

1. $\frac{2a}{(a-3)(a+3)}\times\frac{a-3}{5a}$
2. $\frac{2a}{a+3}\times\frac{1}{5a}$
3. $\frac{2}{a+3}\times\frac{1}{5}$

$$\frac{2}{5(a+3)}$$

Guided Practice

Simplify

a $\frac{3x+6}{5}\times\frac{10}{x+2}$

$=\frac{3(x+2)}{5}\times\frac{10}{x+2}$

$=$

6

b $\frac{3x+3}{2x}\times\frac{x^2}{x^2-1}$

$=\frac{3(\quad)}{2x}\times\frac{x^2}{(\quad)(\quad)}$

$=$

$\frac{3x}{2(x-1)}$

c $\frac{a^2-4}{3}\times\frac{5b}{a+2}$

$\frac{5b(a-2)}{3}$

d $\frac{6x-12}{5}\div\frac{x-2}{3}$

$\frac{18}{5}$

e $\frac{3-6y}{8}\div\frac{1-2y}{2}$

$\frac{3}{4}$

f $\frac{12x}{x+1}\div\frac{6x}{x^2+2x+1}$

$2(x+1)$

Identify lowest common multiple of two algebraic terms

Identify the lowest common multiple of these terms.

a	$6x \ \ \& \ \ 9x^2$	**b**	$(x-2) \ \ \& \ \ 4(x-2)$
c	$(x+3)(x-1) \ \ \& \ \ (x-1)(x+7)$	**d**	$x(x+4) \ \ \& \ \ (x+4)(x-2)$
e	$(x+1)^2 \ \ \& \ \ (x+1)(x-5)$	**f**	$(x+2)(x-3) \ \ \& \ \ (x-3)(2x+1)$
g	$x(x-2) \ \ \& \ \ (x-2)(x+3)$	**h**	$4x(x+5) \ \ \& \ \ 2x^2(x+5)(x-3)$
i	$x+2 \ \ \& \ \ x^2+5x+6$	**j**	$x^2-4 \ \ \& \ \ x+2$

Adding and Subtracting Fractions by Factorising

1. **Factorise** the denominators.
2. Change denominators to the **Lowest Common Denominator** using equivalent fractions.
3. Add or subtract the numerators.

Note: All fractions have invisible brackets in the numerator and denominator $\frac{a+b}{c+d} = \frac{(a+b)}{(c+d)}$

Example Add and subtract algebraic fractions by factorising

$$\frac{1}{x-4}+\frac{1}{x}$$

2. $\frac{}{x(x-4)}+\frac{}{x(x-4)}$

$$\frac{x}{x(x-4)}+\frac{x-4}{x(x-4)}$$

3. $\frac{2x-4}{x(x-4)}$

$$\frac{2+x}{x^2-x}-\frac{5}{x-1}$$

1. $\frac{2+x}{x(x-1)}-\frac{5}{x-1}$

2. $\frac{}{x(x-1)}-\frac{}{x(x-1)}$

$$\frac{2+x}{x(x-1)}-\frac{5x}{x(x-1)}$$

3. $\frac{2-4x}{x(x-1)}$

Guided Practice

Simplify these expressions.

a $\frac{1}{x}+\frac{1}{x+1}$

$=\frac{}{x(x+1)}+\frac{}{x(x+1)}$

$\frac{2x+1}{x(x+1)}$

b $\frac{4}{x+1}+\frac{3}{x-2}$

$\frac{7x-5}{(x+1)(x-2)}$

c $\frac{2}{x-3}-\frac{3}{x-2}$

$\frac{-x+5}{(x-3)(x-2)}$

d $\frac{3}{(x-1)^2}-\frac{2}{x-1}$

$\frac{5-2x}{(x-1)^2}$

e $\frac{2x}{x-5}-\frac{x}{x+1}$

$\frac{x^2+7x}{(x-5)(x+1)}$

f $\frac{2}{x^2-4}-\frac{3}{x+2}$

$\frac{8-3x}{(x-2)(x+2)}$

Compound Fractions

Identify main numerator and main denominator

A compound fraction has a fraction in its numerator, denominator, or both.

Dealing with these fractions requires identifying the ‘main’ numerator and ‘main’ denominator.

a $\dfrac{\frac{1}{x}}{\frac{2}{x+1}}$	**b** $\dfrac{\frac{3}{x}}{1-\frac{5}{x}}$	**c** $\dfrac{1-\frac{2}{x}}{3}$
d $\dfrac{1+\frac{1}{x}}{1-\frac{1}{x}}$	**e** $\dfrac{2+\frac{1}{x+1}}{3-\frac{1}{x+1}}$	**f** $\dfrac{\frac{1}{x}+\frac{1}{y}}{\frac{1}{x}-\frac{1}{y}}$

Simplifying Compound Fractions

1. Identify the main numerator and main denominator.
2. Multiply every single term in the main numerator and denominator by the LCD.
3. Simplify.

This is the fastest method. An alternate method is:

1. Create a single fraction in the main numerator and denominator.
2. Divide the numerator by the denominator.
3. Simplify.

Example Simplify expressions involving compound fractions

$$\frac{1+\frac{1}{3x}}{1-\frac{1}{x^2}} \quad \begin{matrix}\times 3x^2 \\ \hline \times 3x^2\end{matrix}$$

$$= \frac{3x^2\left(1+\frac{1}{3x}\right)}{3x^2\left(1-\frac{1}{x^2}\right)}$$

$$= \frac{3x^2+x}{3x^2-3}$$

$$= \frac{x(3x+1)}{3(x-1)(x+1)}$$

$$\frac{1+\frac{1}{3x}}{1-\frac{1}{x^2}}$$

$$= \frac{\left(\frac{3x+1}{3x}\right)}{\left(\frac{x^2-1}{x^2}\right)} = \frac{3x+1}{3x} \div \frac{x^2-1}{x^2}$$

$$= \frac{3x+1}{3x} \times \frac{x^2}{x^2-1}$$

$$= \frac{x(3x+1)}{3(x-1)(x+1)}$$

Guided Practice

Simplify

a $\frac{1+\frac{1}{x}}{1-\frac{1}{x}}$

$\frac{x+1}{x-1}$

b $\frac{2+\frac{1}{x+1}}{3-\frac{1}{x+1}}$

$\frac{2x+3}{3x+2}$

c $\frac{1-\frac{1}{5x}}{1-\frac{1}{x^2}}$

$\frac{x(5x-1)}{5(x-1)(x+1)}$

d $\frac{\frac{1}{x}+\frac{1}{y}}{\frac{1}{x}-\frac{1}{y}}$

$\frac{x+y}{y-x}$

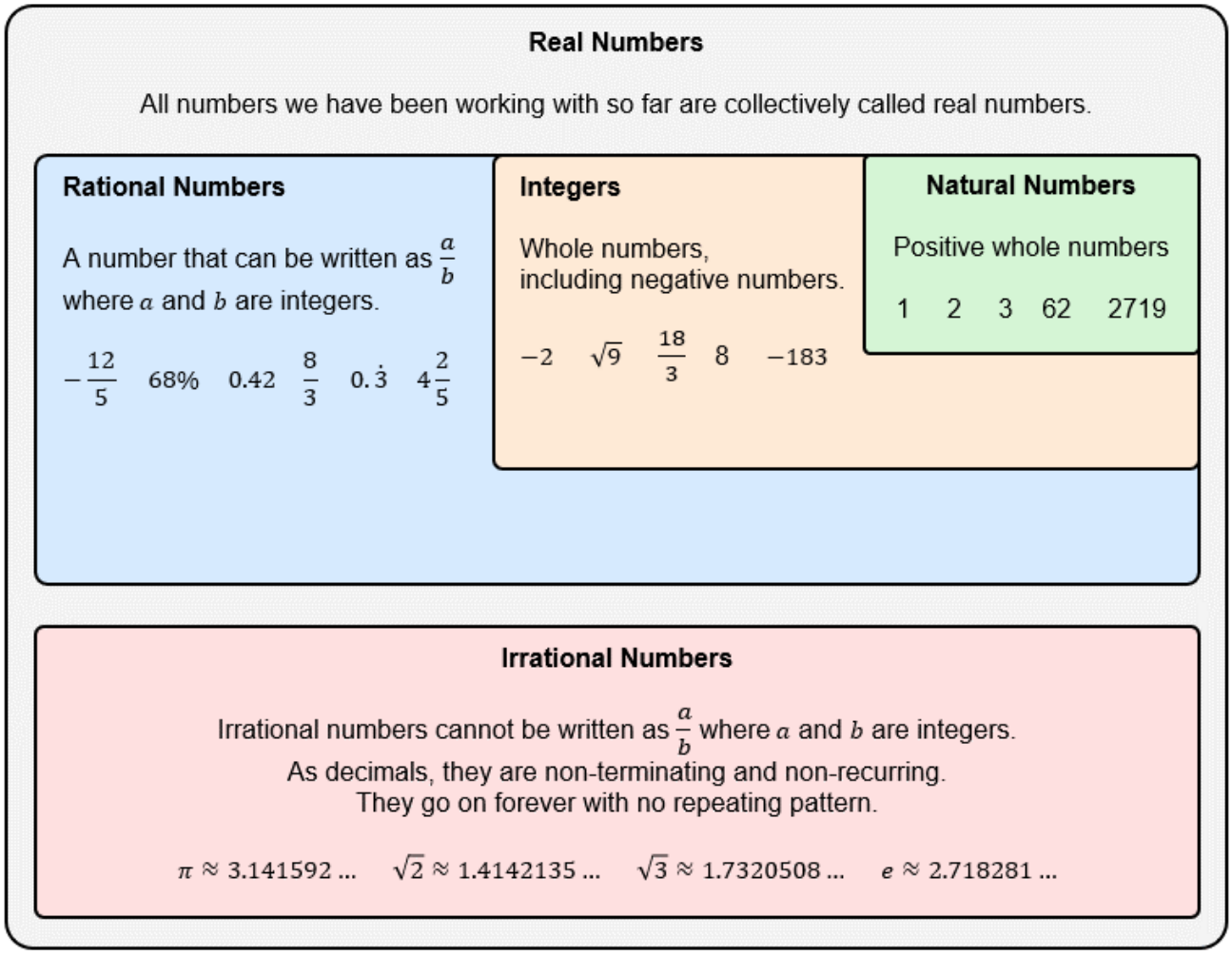
Surds

ⓘ Rational vs Irrational

Rational numbers can be written as a fraction $\frac{a}{b}$ where a and b are integers and $b \neq 0$.

Irrational numbers cannot be written as a fraction, such as π and $\sqrt{2}$.

As decimals, they are **non-terminating** and **non-recurring**.



Identify whether a number is rational or irrational

Using a calculator, represent these numbers as decimals and decide whether they are irrational or rational.

a	$\frac{1}{3}$	Rational	Irrational
b	3	Rational	Irrational
c	0.3	Rational	Irrational
d	$0.\dot{3}$	Rational	Irrational
e	$\sqrt{3}$	Rational	Irrational
f	$\sqrt{9}$	Rational	Irrational
g	0.25	Rational	Irrational
h	25%	Rational	Irrational
i	$\sqrt{4}$	Rational	Irrational
j	$\sqrt{5}$	Rational	Irrational
k	$\sqrt{5}+1$	Rational	Irrational
l	$\frac{\sqrt{2}}{2}$	Rational	Irrational
m	$2\sqrt{2}$	Rational	Irrational
n	π	Rational	Irrational
o	$-\frac{22.5}{7}$	Rational	Irrational
p	$3.\overline{14}$	Rational	Irrational

ⓘ Roots and Surds

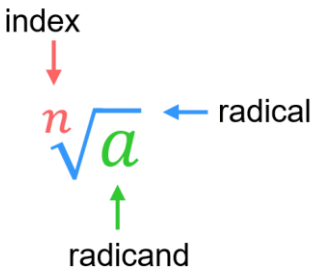
- Square rooting $\sqrt{x}$ is the inverse operation of squaring x^2.
- We call a root a **surd** if it is irrational: E.g., $\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt[3]{7}$
- It is best to leave surds in exact form, rather than a rounded decimal.
- It is impossible (for now) to square root a negative number $\sqrt{-4}$, we say it is **not real**.

🔍 **Identify** surds

Determine whether these roots are surds

a	$\sqrt{3}$	Surd	Rational	Not real
b	$2\sqrt{3}$	Surd	Rational	Not real
c	$\sqrt{4}$	Surd	Rational	Not real
d	$\sqrt{8}$	Surd	Rational	Not real
e	$\sqrt{16}$	Surd	Rational	Not real
f	$2\sqrt{16}$	Surd	Rational	Not real
g	$3+\sqrt{16}$	Surd	Rational	Not real
h	$\sqrt[3]{16}$	Surd	Rational	Not real
i	$\sqrt[3]{8}$	Surd	Rational	Not real
j	$\sqrt[3]{6}$	Surd	Rational	Not real
k	$\sqrt{6}$	Surd	Rational	Not real
l	$\sqrt{9}$	Surd	Rational	Not real
m	$-\sqrt{9}$	Surd	Rational	Not real
n	$\sqrt[3]{9}$	Surd	Rational	Not real
o	$\sqrt{0}$	Surd	Rational	Not real
p	$\sqrt{-0}$	Surd	Rational	Not real
q	$\sqrt{-1}$	Surd	Rational	Not real
r	$-\sqrt{1}$	Surd	Rational	Not real
s	$-\sqrt{-1}$	Surd	Rational	Not real
t	$\sqrt[3]{-8}$	Surd	Rational	Not real
u	$\sqrt{-(-25)}$	Surd	Rational	Not real

ⓘ Language of Surds



Identify parts of a surd

Surd	Index	Radicand
$\sqrt[3]{7}$		
$\sqrt{7}$		
$\sqrt[3]{a}$		
$3\sqrt{a}$		
$\sqrt[9]{n^2}$		
$\sqrt[9]{n+5}$		
$2\sqrt[6]{n^2+5}$		
	5	z^3

ⓘ Surd Laws

Inverse Relationship	Surd Law of Multiplication	Surd Law of Division
$(\sqrt{x})^2 = x$ $\sqrt{x^2} = x$	$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$	$\sqrt{a} \div \sqrt{b} = \sqrt{\frac{a}{b}}$

🔍 Multiply and Divide Surds using the Laws

$(\sqrt{11})^2$ $= 11$	$\sqrt{3} \times \sqrt{10}$ $= \sqrt{30}$	$\sqrt{24} \div \sqrt{8}$ $= \sqrt{3}$

Simplify these expressions.

a $\sqrt{3} \times \sqrt{7}$ $\sqrt{21}$	**b** $\sqrt{8} \times \sqrt{2}$ 4	**c** $\sqrt{2} \times \sqrt{11}$ $\sqrt{22}$	**d** $(\sqrt{12})^2$ 12
e $\sqrt{20} \div \sqrt{2}$ $\sqrt{10}$	**f** $\sqrt{45} \div \sqrt{5}$ 3	**g** $\sqrt{18} \div \sqrt{6}$ $\sqrt{3}$	**h** $\sqrt{49} \div \sqrt{7}$ $\sqrt{7}$

🔍 **Identify** whether you can use a surd law to simplify

a	$\sqrt{7} \times \sqrt{5}$	Yes | No	**e**	$\sqrt{6} \times \sqrt{0.5}$	Yes | No
b	$\sqrt{7} + \sqrt{5}$	Yes | No	**f**	$\sqrt{6} \div \sqrt{10}$	Yes | No
c	$\sqrt{7} \div \sqrt{5}$	Yes | No	**g**	$\sqrt{6} + \sqrt{12}$	Yes | No
d	$\sqrt{7} - \sqrt{7}$	Yes | No	**h**	$\sqrt{6} - \sqrt{3}$	Yes | No

Simplify surds

1. Factorise the **radicand** (number inside the root) according to its **highest square factor**.

   $4,\quad 9,\quad 16,\quad 25,\quad 36,\quad 49,\quad 64,\quad 81,\quad 100,\quad 121,\quad 144\ldots$

2. Split the surd, then simplify perfect squares.

Example Simplify surds

$\sqrt{45}$

1. $\sqrt{9\times 5}$
2. $\sqrt{9}\times\sqrt{5}$

   $=3\sqrt{5}$

$3\sqrt{40}$

1. $3\sqrt{4\times 10}$
2. $3\sqrt{4}\sqrt{10}$

   $=6\sqrt{10}$

$\dfrac{\sqrt{80}}{3}$

1. $\dfrac{\sqrt{16\times 5}}{3}$
2. $\dfrac{4\sqrt{5}}{3}$

Guided Practice

Simplify these surds.

a $\sqrt{12}$ $2\sqrt{3}$

b $\sqrt{28}$ $2\sqrt{7}$

c $\sqrt{54}$ $3\sqrt{6}$

d $3\sqrt{8}$ $6\sqrt{2}$

e $\dfrac{\sqrt{48}}{2}$ $2\sqrt{3}$

f $\dfrac{5\sqrt{40}}{6}$ $\dfrac{5\sqrt{10}}{3}$

g $\sqrt{\dfrac{75}{9}}$ $\dfrac{5}{\sqrt{3}}$ or $\dfrac{5\sqrt{3}}{3}$

h $\sqrt{\dfrac{50}{16}}$ $\dfrac{5}{2\sqrt{2}}$ or $\dfrac{5\sqrt{2}}{4}$

Addition and Subtraction of Surds

We can only add and subtract "**like**" surds; like surds have the same radicand and index.

1. Simplify all surds.
2. Collect like surds.

Example Add and subtract surds

$3\sqrt{7} - \sqrt{6} + 2\sqrt{7}$

2. $3\sqrt{7} - \sqrt{6} + 2\sqrt{7}$

$= 5\sqrt{7} - \sqrt{6}$

$\sqrt{44} + \sqrt{99}$

1. $\sqrt{4}\sqrt{11} + \sqrt{9}\sqrt{11}$

$2\sqrt{11} + 3\sqrt{11}$

2. $5\sqrt{11}$

Guided Practice

Simplify these expressions.

a	$2\sqrt{11} + 6\sqrt{11}$	**b**	$3\sqrt{6} + \sqrt{6}$	**c**	$5\sqrt{7} + 3\sqrt{5} - 4\sqrt{7}$
	$8\sqrt{11}$		$4\sqrt{6}$		$\sqrt{7} + 3\sqrt{5}$
d	$\sqrt{8} + \sqrt{2}$	**e**	$\sqrt{12} - \sqrt{3}$	**f**	$\sqrt{54} + \sqrt{24}$
	$3\sqrt{2}$		$\sqrt{3}$		$5\sqrt{6}$
g	$2\sqrt{8} - \sqrt{18}$	**h**	$2\sqrt{5} + \sqrt{45}$	**i**	$\sqrt{72} - \sqrt{50} + \sqrt{12}$
	$\sqrt{2}$		$5\sqrt{5}$		$2\sqrt{3} + \sqrt{2}$

Multiplying Surds

$$a\sqrt{x} \times b\sqrt{y} = ab\sqrt{xy}$$

1. Multiply the coefficients and the radicands separately.
2. Simplify, if possible.

Example Multiply surds

$\sqrt{15} \times \sqrt{5}$	$5\sqrt{6} \times 7\sqrt{10}$
1. $\sqrt{75}$	1. $35\sqrt{60}$
2. $5\sqrt{3}$	2. $70\sqrt{15}$

Guided Practice

Simplify these expressions.

a $\sqrt{7} \times \sqrt{3}$ $\sqrt{21}$	**b** $\sqrt{5} \times \sqrt{20}$ 10	**c** $\sqrt{12} \times \sqrt{8}$ $4\sqrt{6}$
d $2 \times 3\sqrt{2}$ $6\sqrt{2}$	**e** $2\sqrt{6} \times 5\sqrt{3}$ $30\sqrt{2}$	**f** $-4\sqrt{5} \times 3\sqrt{11}$ $-12\sqrt{55}$
g $2\sqrt{3} \times 3\sqrt{15}$ $18\sqrt{5}$	**h** $(2\sqrt{5})^2$ 20	**i** $3\sqrt{2} \times -4\sqrt{6}$ $-24\sqrt{3}$

Dividing Surds

$$a\sqrt{x} \div b\sqrt{y} = \frac{a}{b}\sqrt{\frac{x}{y}}$$

1. Divide the coefficients and the radicands separately.
2. Simplify, if possible.

Example Divide surds

$\sqrt{10} \div \sqrt{15}$

1. $\sqrt{\frac{10}{15}}$
2. $= \sqrt{\frac{2}{3}}$

   $= \frac{\sqrt{2}}{\sqrt{3}}$

$18\sqrt{48} \div 6\sqrt{2}$

1. $\frac{18}{6}\sqrt{\frac{48}{2}}$
2. $= 3\sqrt{24}$

   $= 3\sqrt{4}\sqrt{6}$

   $= 6\sqrt{6}$

Guided Practice

Simplify:

a $\sqrt{15} \div \sqrt{3}$ $\sqrt{5}$

b $3\sqrt{10} \div \sqrt{5}$ $3\sqrt{2}$

c $6\sqrt{21} \div 3\sqrt{7}$ $2\sqrt{3}$

d $8\sqrt{11} \div 6\sqrt{25}$ $\frac{4}{15}\sqrt{11}$

e $3\sqrt{80} \div 8\sqrt{5}$ $\frac{3}{2}$

f $\sqrt{2\frac{1}{4}}$ $\frac{3}{2}$

Expanding Brackets with Surds

$$a(b+c) = ab + ac$$

1. Expand brackets.
2. Simplify.

Example Expand brackets with surds

$\sqrt{3}(3\sqrt{5}+\sqrt{6})$	$3\sqrt{6}(2\sqrt{10}-4\sqrt{6})$
1. $3\sqrt{15}+\sqrt{18}$	1. $6\sqrt{60}-12\sqrt{36}$
2. $3\sqrt{15}+3\sqrt{2}$	2. $12\sqrt{15}-72$

Guided Practice

Simplify:

a $\sqrt{2}(\sqrt{10}-\sqrt{2})$ $2\sqrt{5}-2$	**b** $\sqrt{6}(\sqrt{8}-2)$ $4\sqrt{3}-2\sqrt{6}$	**c** $\sqrt{3}(\sqrt{2}+\sqrt{5})$ $\sqrt{6}+\sqrt{15}$
d $2\sqrt{3}(\sqrt{5}+\sqrt{7})$ $2\sqrt{15}+2\sqrt{21}$	**e** $3\sqrt{2}(2\sqrt{13}-\sqrt{11})$ $6\sqrt{26}-3\sqrt{22}$	**f** $4\sqrt{5}(\sqrt{5}-\sqrt{10})$ $20-20\sqrt{2}$

Expanding Brackets with Surds

$$(a+b)(c+d) = ac + ad + bc + bd$$

1. Expand brackets.
2. Simplify.

Example Expand brackets with surds

$(\sqrt{15}+2)(\sqrt{3}-3)$

$= \sqrt{45} - 3\sqrt{15} + 2\sqrt{3} - 6$

$= 3\sqrt{5} - 3\sqrt{15} + 2\sqrt{3} - 6$

$(\sqrt{15}-\sqrt{6})^2$

$= 15 - 2\sqrt{90} + 6$

$= 21 - 2\sqrt{90}$

$= 21 - 6\sqrt{10}$

Guided Practice

Simplify:

a $(\sqrt{3}+1)(\sqrt{2}-1)$

$\sqrt{6} - \sqrt{3} + \sqrt{2} - 1$

b $(\sqrt{5}+1)(\sqrt{5}-1)$

4

c $(\sqrt{7}-\sqrt{5})^2$

$12 - 2\sqrt{35}$

d $(2\sqrt{6}-1)(\sqrt{6}-2)$

$14 - 5\sqrt{6}$

e $(5\sqrt{2}-1)(3\sqrt{2}+3)$

$27 + 12\sqrt{2}$

f $(\sqrt{13}+\sqrt{9})^2$

$22 + 6\sqrt{13}$

Rationalising the Denominator

Rationalising the Denominator

- Rationalising the denominator means to write a fraction with an integer denominator.
- Multiply the numerator and denominator by the surd in the denominator.

$\frac{a}{\sqrt{b}} \times \frac{\sqrt{b}}{\sqrt{b}} = \frac{a\sqrt{b}}{b}$	$\frac{a}{b\sqrt{c}} \times \frac{\sqrt{c}}{\sqrt{c}} = \frac{a\sqrt{c}}{bc}$	$\frac{a\sqrt{b}}{c\sqrt{d}} \times \frac{\sqrt{d}}{\sqrt{d}} = \frac{a\sqrt{bd}}{cd}$

Example Success Criteria

$\frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$

$= \frac{\sqrt{2}}{2}$

$\frac{2}{5\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$

$= \frac{2\sqrt{3}}{15}$

Guided Practice

Rationalise these expressions.

a $\frac{1}{\sqrt{3}}$

$\frac{\sqrt{3}}{3}$

b $\frac{\sqrt{2}}{\sqrt{3}}$

$\frac{\sqrt{6}}{3}$

c $\frac{1+\sqrt{2}}{\sqrt{3}}$

$\frac{\sqrt{3}+\sqrt{6}}{3}$

d $\frac{\sqrt{6}-5}{\sqrt{2}}$

$\frac{2\sqrt{3}-5\sqrt{2}}{2}$

e $\frac{\sqrt{3}}{2\sqrt{2}}$

$\frac{\sqrt{6}}{4}$

f $\frac{8+3\sqrt{2}}{4\sqrt{5}}$

$\frac{8\sqrt{5}+3\sqrt{10}}{20}$

ⓘ The Conjugate

The **conjugate** of $a + b\sqrt{c}$ is $a - b\sqrt{c}$.

🔍 **Identify** the conjugate

Expression	Conjugate
$7 + \sqrt{3}$	
$8 + \sqrt{5}$	
$\sqrt{5} + 8$	
$\sqrt{5} - 8$	
$4 - \sqrt{7}$	
$5 + 3\sqrt{2}$	
$6 - 2\sqrt{3}$	
$4\sqrt{6} - 3$	

ⓘ The Conjugate Product

- The product of an expression and its conjugate is a difference of two squares.
- A conjugate product has no surds.

$(a + \sqrt{c})(a - \sqrt{c})$ $= a^2 - c$	$(a + b\sqrt{c})(a - b\sqrt{c})$ $= a^2 - b^2c$

🔍 **Identify** the conjugate product

a $(2 + \sqrt{3})(2 - \sqrt{3})$

$= 2^2 - (\sqrt{3})^2$

$= 4 - 3$

$= 1$

b $(5 - \sqrt{6})(5 + \sqrt{6})$

19

c $(7 - \sqrt{3})(7 + \sqrt{3})$

46

d $(5 + 3\sqrt{6})(5 - 3\sqrt{6})$

−29

e $(7 + 2\sqrt{5})(7 - 2\sqrt{5})$

29

f $(12 - 2\sqrt{3})(12 + \sqrt{3})$

132

Rationalising Binomial Denominator

For binomial denominators, rationalise by **multiplying by the conjugate**.

$$\frac{a}{\sqrt{b}+\sqrt{c}} \times \frac{\sqrt{b}-\sqrt{c}}{\sqrt{b}-\sqrt{c}} = \frac{a\sqrt{b}-a\sqrt{c}}{b^2-c^2}$$

$$\frac{a}{\sqrt{b}-\sqrt{c}} \times \frac{\sqrt{b}+\sqrt{c}}{\sqrt{b}+\sqrt{c}} = \frac{a\sqrt{b}+a\sqrt{c}}{b^2-c^2}$$

Example Rationalise a binomial denominator

$$\frac{3}{5+\sqrt{3}} \times \frac{5-\sqrt{3}}{5-\sqrt{3}}$$

$$= \frac{15-3\sqrt{3}}{25-3}$$

$$= \frac{15-3\sqrt{3}}{22}$$

$$\frac{\sqrt{5}}{2\sqrt{2}-3} \times \frac{2\sqrt{2}+3}{2\sqrt{2}+3}$$

$$= \frac{2\sqrt{10}+3\sqrt{5}}{4\times 2-9}$$

$$= -\left(2\sqrt{10}+3\sqrt{5}\right)$$

Guided Practice

Rationalise these expressions.

a $\frac{1}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1}$

$\frac{\sqrt{3}+1}{2}$

b $\frac{1}{\sqrt{5}-\sqrt{2}}$

$\frac{\sqrt{5}+\sqrt{2}}{3}$

c $\frac{4}{\sqrt{3}+\sqrt{2}}$

$4\sqrt{3}-4\sqrt{2}$

d $\frac{\sqrt{3}-4}{\sqrt{3}+4}$

$\frac{-19+8\sqrt{3}}{13}$

e $\frac{2\sqrt{7}}{2\sqrt{7}-5}$

$\frac{28+10\sqrt{7}}{3}$

f $\frac{2\sqrt{3}}{\sqrt{5}+2\sqrt{6}}$

$\frac{-2\sqrt{15}+12\sqrt{2}}{19}$